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AI Reservoir Characterization

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The accurate delineation of hydrocarbon-bearing intervals is essential for efficient reservoir development, influencing decisions ranging from well placement to production optimization and resource evaluation. Traditional workflows employ seismic inversion to derive elastic-property volumes such as acoustic impedance (I_p) and the V_p/V_s ratio, followed by rock-physics-template interpretation to distinguish fluid effects (Ødegaard and Avseth, 2004). Although these methods are widely adopted, they introduce uncertainties at multiple stages. Inversion quality depends on accurate wavelet estimation, both in frequency content and phase alignment (Latimer et al., 2000), constructing the low-frequency background model often relies on sparsely spaced well logs (Sams and Carter, 2017) and assumptions of weak lithology contrasts (Avseth et al., 2016) or isotropy can neglect anisotropy and rock heterogeneity (Asaka, 2018). These factors undermine confidence in fluid predictions and risk costly misinterpretations.

Artificial Intelligence (AI), and specifically Deep Learning (DL) circumvents many inversion pitfalls by learning direct mappings from seismic and geological inputs to subsurface properties. Convolutional neural networks (CNNs) can learn complex, non-linear mappings between seismic amplitudes and target attributes, capturing geological and geophysical context if appropriately trained (Zhang et al., 2021; Li et al., 2022). Such approaches can bypass explicit inversion approaches by directly inferring elastic or lithologic parameters from seismic and ancillary data. Synthetic studies suggest that deep learning can predict porosity, permeability, and elastic moduli with high accuracy under idealized conditions. Yet practical deployment for field-scale fluid detection has been limited by challenges in generalization, scarcity of representative training data, and difficulty in integrating with rock-physics theory.

Heir and Aghayev (2025) started bridging the gap by applying a 1D Temporal Convolutional Network (TCN) to predict V_p , V_s , and I_p at well locations, and subsequently applying rock physics, demonstrating the important contribution of seismic data for DL prediction of blind-well estimates of fluid presence. However, their approach remained limited to individual well traces.

Introduction

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In this study, we extend the methodology to full 3D volume inference and comprehensive validation. We present an end-to-end deep-learning workflow for fluid prediction that integrates geological context, rock-physics constraints, and data-driven inference. We apply a one-dimensional CNN, inspired by TCN architectures, to predict V_p/V_s ratio and I_p from 3D seismic data, velocity, Relative Geologic Time (RGT), and discrete stratigraphic units, building on the stratigraphy-guided deep learning method of Heir et al. (2024). The resulting elastic volumes are then plotted on an I_p -versus- V_p/V_s RPT, where deviations below the brine-sandstone trend, derived from Dvorkin–Nur contact-cement modeling and Hashin–Shtrikman bounds, are interpreted as hydrocarbon indicators (Avseth et al., 2005).

To evaluate performance, we benchmark the DL+RPT workflow against distance-weighted interpolation guided by RGT (Yan et al., 2022) and physics-based elastic inversion (Huy, 2019). Validation is performed on inline 10210 using wells F-1B (blind), F-11A, F-11T2, and F-1A. Evaluation includes qualitative assessment via cross-plot coherence and inline fluid-presence maps, as well as quantitative metrics such as Pearson correlation, mean absolute error (MAE), and F1 classification score at a blind well.

Our key contributions are (1) the introduction of a field-validated DL+RPT workflow for seismic-based hydrocarbon detection, (2) a quantitative comparison against state-of-the-art interpolation and inversion baselines, and (3) demonstration of high-accuracy fluid prediction ($F1 = 0.78$) at a blind well in a mature producing field. The remainder of this manuscript describes the geological setting and data, details of the DL architecture and RPT methodology, results and comparisons, discussion of implications, and concluding remarks.

Geological Setting and Data

The Volve field occupies block 15/9 in the southern Viking Graben, approximately 160 km southwest of Stavanger, Norway, at a water depth of 80 meters. Operators discovered Volve in 1993, and it produced hydrocarbons from the Hugin Formation of the Middle Jurassic age at 2700–3100 meters depth (Norwegian Offshore Directorate, 2025). The reservoir consists of clean, quartz-rich sandstones with porosities of 18–24 % (Otoo and Hodgetts, 2021) and net-to-gross ratio of 0.9 on average (Yan et al., 2022). Hydrocarbons are trapped in tilted fault-block structures sealed by the overlying Draupne shale.

The wells used in this study include 12 wells with measured I_p and V_p , and a subset of 7 wells also includes V_s measurements. One well, 15/-9-F-1B, is reserved as a blind well for validation, while the remaining wells are used for training the deep learning model. For testing, we focus on inline 10210, as it intersects four key wells - 15/9-F-1A, F-11A, F-11T2, and the blind well 15/9-F-1B. The inline has also been used for previous studies (Huy, 2019). The four wells all have measurements of V_p , I_p and V_s . Although these deviated wellbores project onto a single inline, the surrounding subsurface exhibits only mild lateral heterogeneity, so the projection-induced errors are expected to be negligible.

The seismic dataset consists of the Kirchhoff time-migrated Prestack Depth Migration (PSDM) ST100110ZC11 survey, with full, near, mid, and far angle stacks covering angles up to 41° . It covers a $5 \text{ km} \times 5 \text{ km}$ area with $12.5 \text{ m} \times 12.5 \text{ m}$ bin spacing and 4 ms sampling. The source wavelet frequency peaks at 30 Hz and the Hugin reservoir dominant frequency is 20 Hz. We do not condition the seismic or do other changes to the migrated seismic images. In addition to the amplitude volume, we used the RMS stacking velocity field and converted it to interval velocity, which represents the average seismic-wave velocity within specific geological strata.

Equinor has picked some key stratigraphic horizons throughout the seismic volume: Ty, Shetland, Base Cretaceous Unconformity (BCU), Hugin top, and Hugin base. We use these horizons to provide the structural framework for generating a volume of stratigraphic units. Stratigraphic units delineate formation boundaries and provide categorical masks that enable the network to distinguish layer-specific elastic responses. The horizons also serve as surfaces for constructing Relative Geologic Time (RGT) models. The geological age associated with each horizon is derived from chronostratigraphic charts, and the surfaces are then used to interpolate RGT throughout the seismic volume. The RGT volume established an age continuum, normalizing for structural dip and variations in layer thickness, so that units deposited during the same geological age retain an identical age value regardless of their current depth. Collectively, these discrete and continuous geological descriptors imbue the model with robust spatial constraints, yielding elastic-property predictions that are both geologically coherent and temporally consistent.

We established well-to-seismic ties by first anchoring the time-depth framework with available checkshots measurements. Building on this foundation, we then employed a partially automated auto-tie workflow - using Tscannan et al's (2022) deep learning and Bayesian search algorithm. It generates synthetic seismograms from well logs and correlates them with seismic traces - thereby improving calibration for wells lacking checkshot measurements or relying on approximate time-depth estimates. Finally, to further refine the model and ensure robust geological consistency, we correlated interpreted seismic horizons with well-tops picks, sharpening the time-depth calibration and reinforcing the alignment between stratigraphic markers and subsurface seismic reflections.

Wells logs are low-pass filtered to approximate the seismic bandwidth, removing high-frequency variations beyond what the seismic wavelet can resolve. Afterwards, we resampled the elastic logs at 1 ms, and then scaled them with a RobustScaler, which centers on the median and scales by the interquartile range to minimize the impact of outliers.

Methodology

We implement a three-stage workflow: (1) deep-learning-based prediction of elastic properties, (2) RPT fluid discrimination, and (3) benchmarking against traditional methods.

Deep Learning Framework

The method employs a supervised deep learning approach using a TCN (Bai et al, 2018) to learn the relationship between input features and target elastic properties by training at well locations. The input features include seismic data and stratigraphic information - both discrete and continuous - such as Relative Geologic Time (RGT). This ensures that the model incorporates both geological and geophysical context.

Each network is trained independently for one target property, with only one channel per model. After training, the model is applied to the full 3D volume to produce field-wide predictions.

The TCN is configured with a kernel size of 4 and 128 filters per layer, allowing it to effectively capture local patterns in the input sequence. It uses exponentially increasing dilation rates to expand the receptive field and model both short- and long-term dependencies without increasing model depth. Residual skip connections facilitate gradient flow and training convergence, while ReLU activation functions provide non-linear transformations with each block. This configuration optimizes the model's ability to learn mapping from seismic data and geological context to target properties measured at wells.

Training and Inference

During training, each target measurement is paired one-to-one with its corresponding input feature vector at the same depth point, ensuring a direct correspondence between input and measured properties. The model parameters are optimized to minimize the mean squared error between the TCN's predictions and the measured target property at well locations. Optimization is performed using the Adam algorithm, which adaptively adjusts individual learning rates to facilitate faster and more stable convergence. To mitigate overfitting, dropout layers are introduced, and training is terminated via early stopping once the validation loss fails to improve over a specified patience. Feature selection ensures that the features are contributing positively to the prediction accuracy.

We train on paired well-log and volume feature samples along the trajectories from ten wells, using one well for validation, and withholding F-1B for blind evaluation. After the validation loss stabilized, the best-performing model checkpoint was selected for full 3D volume inference. The blind well, F-1B, was entirely withheld until post-training evaluation.

We then inferred the trained model on the entire volume, trace by trace, producing continuous volumes of predicted V_p/V_s and I_p . These volumes were then subjected to the following rock-physics workflow.

Time

The training time was under a minute, and the prediction time was a few seconds.

Rock-Physics Template Construction and Fluid Discrimination

The fluid-detection stage projects predicted elastic properties onto an I_p versus V_p/V_s cross-plot. We computed a brine-sandstone trend line using the Dvorkin–Nur contact-cement model, parameterized for Volve reservoir conditions. The Hashin–Shtrikman upper bounds for brine saturation define the maximum elastic response for water-saturated sandstones. Samples plotting below this line are flagged as hydrocarbon-bearing, while those above are considered brine-bearing. This RPT approach enables direct fluid classification without requiring additional inversion steps.

Benchmark Methods

To contextualize performance, we compared the DL+RPT workflow against two established methods. The first, distance-weighted interpolation, uses a radial basis-function interpolator guided by RGT distances to distribute well-derived elastic logs across the volume (Yan et al., 2022). While it honors stratigraphic continuity, the smoothing inherent to interpolation often blurs discrete fluid contacts. The second benchmark is a physics-based elastic inversion following Huy (2019), which estimates near- and far-angle impedances using a statistical wavelet and a low-frequency model constructed from interpolated well logs. Fluid cutoffs determined from training wells are then applied volumetrically for classification.

Evaluation Metrics

We assessed workflow performance qualitatively through RPT cross-plots and fluid-presence maps on inline 10210, and quantitatively using three metrics at well locations. Pearson correlation coefficients measure linear agreement between predicted and measured elastic attributes. Mean absolute error (MAE) quantifies

average prediction deviation. Finally, fluid-classification accuracy at the blind well is summarized by precision, recall, and F1 score, calculated by comparing RPT-based oil flags of the measured logs with RPT-based flags of the predictions. The F1 score is a measure of classification accuracy that balances precision (the fraction of predicted positives that are correct) and recall (the fraction of actual positives that are detected). A high F1 score indicates that the method both identifies most true positives and minimizes false positives.

Results

Elastic Property Prediction

We first predicted elastic properties in 3D, and Figure 1 shows the I_p prediction. At the three training wells, the DL model achieved high fidelity in elastic-property estimation, with Pearson correlations of 0.92 for I_p and 0.89 for V_p/V_s . Performance on the blind well remained robust, yielding correlations of 0.85 (I_p) and 0.75 (V_p/V_s). In comparison, distance-weighted interpolation produced correlations of 0.56/0.86. The high 0.86 V_p/V_s correlation of the interpolation is typical of interpolation, as it will often successfully describe the background trend. However, there is no accounting for the seismic and the physics, and since it shows a statistical interpolation based on the surrounding wells, including dry wells, the V_p/V_s values in the Hugin reservoir are higher than the measurement, erroneously implying that the well is dry. This becomes evident in the following cross plot analysis, illustrated with Figure 2 (Heir and Aghayev, unpubl. results).

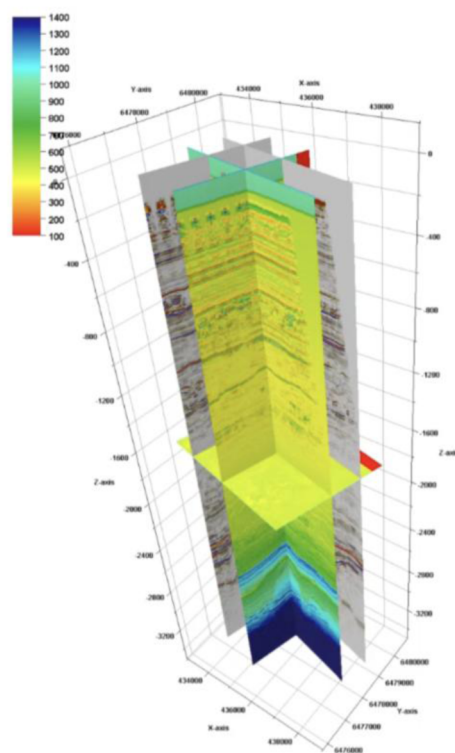


Figure 1—3D prediction of I_p

RPT Cross-Plot Analysis

Figure 2a displays measured well-log points plotted on the I_p – V_p/V_s RPT for the Hugin reservoir interval. The measured samples of the oil-bearing wells F-1B and F-11A cluster below the brine-sandstone trend. Figure 2b shows DL-predicted points replicating this clustering, with a high F1 classification score of 0.85.

By contrast, [Figure 2c](#) illustrates that interpolation misplaces F-1B samples above the brine line, incorrectly classifying the blind well as water-bearing.

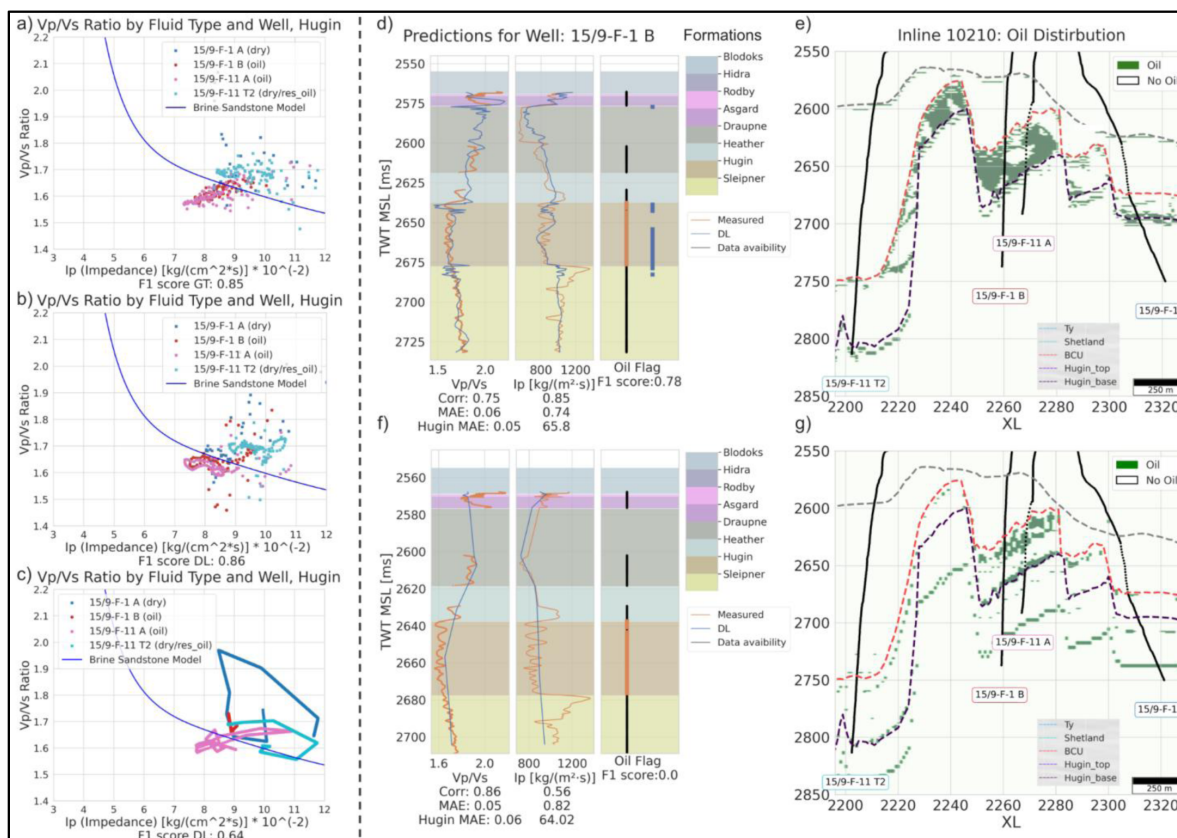


Figure 2—To the left of the dashed line are crossplots of Vp/Vs versus Ip for the Hugin formation showing a) measured log b) DL predictions, and c) interpolated results. To the right of the dashed line are DL predictions for the blind well (d) and inline (e) and the same for interpolation (f, g).

Fluid-Presence Comparisons

[Figure 2d-g](#) compares 1D and cross section fluid-presence for the DL prediction and interpolation. [Figure 2d](#) demonstrates that DL successfully predicts the oil-presence in the F1B blind well, with a high F1 correlation score of 0.78. In contrast, with geostatistical interpolation (1f), no oil is indicated and F1 score is 0.0. The DL +RPT map ([Fig. 1e](#)) reveals a coherent hydrocarbon plume within the Hugin reservoir, extending between wells F-11A and F-1B. The interpolation map ([Fig. 1g](#)) is overly smooth, failing to capture the plume's continuity and missing discrete oil pockets. In comparison, the physics-based fluid estimate ([Huy, 2019](#)) identifies major oil zones but also shows spurious "streaks" of hydrocarbons in the Draupne formation above F-11A, inconsistent with well data and field geology.

Discussion

The results demonstrate that deep-learning-predicted elastic volumes, when interpreted through a rock-physics template, offer a reliable and accurate method for hydrocarbon detection. The high correlation and low MAE in elastic predictions indicate that the TCN has successfully learned the complex relationships between seismic amplitudes, stratigraphic context, and elastic responses. By embedding RGT and geological context into the network inputs, we ensured geological plausibility and honored stratigraphic boundaries, which is critical for coherent fluid maps. The quick training and prediction times enable quick workflows that can shorten turnaround times in field development.

AI reservoir characterization can thus present a valuable complement to conventional methodologies. Compared with distance-weighted interpolation, the DL workflow clearly outperformed in detecting discrete hydrocarbon zones. While DL distinguished the oil sands, interpolation smoothing obscured the sharp fluid contacts. When contrasted with physics-based elastic inversion, the DL+RPT approach was shown to avoid false positives in non-reservoir intervals, a common pitfall arising from misestimation of wavelets or low-frequency models. Inversion remains a valuable tool to understand the physics, but a combined workflow with data-driven methods can ensure an equally good understanding of field-specific nuances that can be read directly from the data.

Nevertheless, the approach has limitations. Deep learning models require representative, high-quality training data; in fields with sparse well control or poor log quality, performance may degrade. Generalization to other basins or lithofacies will likely necessitate transfer learning or retraining with local data. Additionally, the workflow as presented here does not quantify uncertainty. Future work should incorporate Bayesian neural networks or Monte Carlo dropout to estimate prediction confidence, thereby informing risk-based decision making.

Operationally, the DL+RPT workflow offers significant advantages in turnaround time, producing elastic and fluid maps in minutes compared to hours or days for traditional inversion. Its reproducibility and minimal expert tuning facilitate integration into reservoir characterization workflows, enabling rapid scenario testing and real-time interpretation updates. We envision hybrid workflows that harness both DL and physics-based inversion: rapid DL predictions can guide inversion initialization, and targeted inversion can refine critical intervals.

Conclusions

This study introduces a robust, field-validated workflow combining deep learning and rock-physics templates for seismic-based hydrocarbon detection. The proposed CNN accurately predicts Vp/Vs ratio and acoustic impedance, achieving Pearson correlations exceeding 0.85 on a blind well. Subsequent RPT-based fluid classification delivers an F1 score of 0.78, outperforming distance-weighted interpolation (F1 = 0) and physics-based inversion. By integrating geological context through RGT and formation indicators, the workflow produces geologically consistent fluid maps while reducing false positives. The rapid, data-driven approach holds promise for accelerating reservoir characterization in oil and gas, carbon-storage, and geothermal applications. Future research should extend the workflow to diverse reservoirs and embed uncertainty quantification to support decision-making under risk.

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